

ONE MONTH INTERNSHIP PROGRAM ON **DRONE** TECHNOLOGY

PROGRAM DURATION
1 Month

BATCH SIZE:
30 Students



LEARN • BUILD • FLY • INNOVATE

PROGRAM OVERVIEW

This 1-month intensive internship program is designed to provide students with practical exposure, industry-relevant skills, and hands-on experience in emerging Drone technologies.

The program focuses on learning by doing, where students will work on real hardware, live demonstrations, and guided project development.

By the end of the program, students complete their working project, understand industry best practices, and develop the confidence to prototype, implement, and present technology solutions relevant to modern automation and smart systems.

This internship not only enhances technical competence but also nurtures creativity, teamwork, documentation skills, and professional readiness, preparing for careers and higher studies in cutting-edge technology domains.

Module 1.

Fundamentals & Live Demonstrations

A Foundational Exploration through Multi-Platform Live Demonstrations

Introduction to Drone Technology

- ▀ A brief history of Quad Copters.
- ▀ Multi-rotor drones.
- ▀ Demonstration of different series of Drones
 - » F330
 - » F450
 - » Q250
 - » FPV
 - » S500
 - » S550
 - » Tarot series
 - » Mark4 series
- ▀ Applications in industries like healthcare, defence, Surveillance, and Delivery.

DGCA Rules to Fly Custom Drones in India

- ▀ Here, we will learn about the summary of the DGCA rules for flying custom drones in India under the current regulatory framework (based mainly on the Drone Rules 2021 and 2025-26 guidance by the Directorate General of Civil Aviation):

Understanding the concept of Flying zones

- » Green Zone
- » Orange Zone
- » Red Zone

Understanding the working principle of Drone technology

- » The physics behind the working of a drone
- » Primary Forces for Drone Flight
- » Understanding the aerodynamics
- » Working of a drone

Module 2.

Learn about all the basic components

Understanding these core parts helps you design, assemble, troubleshoot, and operate drones effectively.

- ▀ Frames
 - » Types & Materials
- ▀ Brushless motors
 - » How Does a BLDC Motor Works?

- ▀ Propellers
 - » Presentation
 - » Characteristics
- ▀ LiPo (Lithium Polymer) Battery
 - » Voltage Check
 - » Charging & Discharging

ESC (Electronic Speed Controller)

How the ESC is the critical link between the flight controller and the brushless motor?

Understanding the core concept to control 3 phase motors with a DC power supply.

- ▀ **ESC: Electronic Speed Controller**
 - » Inputs and Outputs
 - » Voltage & Outputs
- ▀ **PDB: Power Distribution Board**
 - » Presentation
 - » Technical Parameters
 - » Features
- ▀ **Hands-on Activity**
 - » Binding Transmitter – Receiver pair of FSi6 Radio control system
 - » Binding Transmitter – Receiver pair of Radio link T8Fb Radio control system
 - » Testing the Speed of BLDC Motor
 - » Imax B6 AC charger setup/Configuration
 - » Balance charging of LIPO Battery
 - » Propeller Thrust check
 - » Propeller Balancing

Module 3.

Flight Controller

- ▀ Why the flight controller is the core intelligence of a drone? How does it read sensors, runs control algorithms, and commands the motors hundreds of times per second to keep the aircraft stable and responsive?
- ▀ FC: Flight Controller
- ▀ Processors
- ▀ Must-Have Components (Important accessories for FC)
- ▀ Different Versions of Flight Controller
 - » APM 2.8 Ardupilot
 - » Pixhawk 2.4.8
 - » Radio link Cross Flight Controller

Radio Links (Radio communication Links, The Transmission & Reception System Through Wireless Radio Frequency Channels.)

- ▀ A radio link is the wireless bridge between the pilot, the drone, and (optionally) the ground station. Reliable radio links are essential for control, telemetry, and safety.
 - » RC Transmitter
 - » RC Receiver

Module 4.

Assembly of a Drone (Hands-on Activity)

- ▀ Drone assembly involves systematically integrating the frame, motors, ESCs, flight controller, power system, radio receiver, and sensors into a balanced and vibration-free structure. Proper wiring, secure mounting, correct motor direction, and clean power distribution are critical to ensure stable flight.
- ▀ **Careless and Faulty assembly can lead to a crash and sever damages.**
 - » Frame Assembly
 - » Base fitting.
 - » Arm installation.
 - » Motor Installation.
 - » Receiver Mounting.
 - » FC installation.
 - » Camera Installation.
 - » GPS Installation.
 - » Propeller Mounting

Module 5.

Remote Setup (Hands-on Activity)

- ▀ Setting up an RC remote for a drone involves binding the transmitter with the receiver, connecting the receiver to the flight controller, and configuring the correct protocol (SBUS/iBUS/CRSF). Finally, configure and test the failsafe to ensure the drone responds safely in case of signal loss.
- ▀ Bind the Transmitter and Receiver
- ▀ Connect the Receiver to the Flight Controller
 - » SBUS / iBUS / CRSF / PPM signal wire → FC RX pin
- ▀ Power-up the Receiver
- ▀ Configure Receiver Type in Flight Software
- ▀ Channel Mapping and Stick Calibration
 - » Roll → Aileron
 - » Pitch → Elevator

- » Throttle → Throttle
- » Yaw → Rudder
- ▀ Assign AUX Switches (Flight Modes)
- ▀ Set Failsafe Behaviour
- ▀ Pre-Arm Safety Checks

Module 6.

Programming The FC (Flight Controller)

Installing the Mission Planner ArduPilot software

- ▀ Programming FC
 - » Uploading the Firmware to FC
 - » Setting up the initial tune parameters
 - » Frame selection
 - » Accel Calibration
 - » Simple Accel Calibration
 - » Compass Calibration
 - » Radio Calibration
 - » Battery Monitoring
 - » Flight Modes Setup
 - » Fail Safe Setup

Module 7:

ESC Calibration (Hands-on Activity)

- ▀ The most important task in drone development is the Esc calibration to ensure the proper responses from the motor on each action control.
 - » Esc Manual Calibration
 - » Dry Testing of a Drone
 - » Motor test through the Mission Planner Software
 - » Learning Arming/Disarming
 - » Learning How to Use Different Flight Modes

Module 8.

Simulation Training

- ▀ Learning the Basic controls of Drone Flying through a Simulator Tool.
 - » Roll → Aileron
 - » Pitch → Elevator
 - » Throttle → Throttle
 - » Yaw → Rudder
- ▀ On-ground Real-Time Flight Training (1st Session)

Module 9.

- ▀ On-ground Real-Time Flight Training (2nd Session)
 - » Mounting Lipo Battery
 - » Mounting Propellers
 - » ARMING & real time flight Training.

Module 10.

- ▀ On-ground Real-Time Flight Training (3rd Session)
- ▀ Problem and statement (Onsite assignment)
- ▀ Debugging & Troubleshooting
- ▀ GPS Error Troubleshooting
- ▀ Flight Stability testing
- ▀ Enabling and testing the RTL (Return to Location) feature with GPS

Learning Outcomes

By the end of the internship, students will:

- ▀ Understand core concepts of Drone Technology
- ▀ Gain practical experience with Flight Controller, Radio Transmitter, GPS, and other Components
- ▀ Build and program real-world Drone
- ▀ Develop problem-solving and technical skills
- ▀ Get exposure to industry-level technologies

Certification

- ▀ Internship Completion Certificate
- ▀ Project Participation Certificate

Who Can Join?

- ▀ Diploma / B.Tech / M.Tech Students
- ▀ BCA / MCA / BSc IT Students
- ▀ Any technical background learner interested in Drone development & Aviation

Workshop Methodology

1. On-screen explanation of components, circuits, and programs through a projector.
2. Trainers will be assigned to help participants through real-time development.
3. Workshop Execution, Students' Interaction, Q&A session.
4. Students must carry their own Laptop during the training session.

Enrolment & Payment Guidelines

1. Enrolment confirmation (by institution/student) must be completed at least 10 days prior to the program commencement.
2. Payment is required in advance to confirm participation and reserve the slot.
3. Seats are limited and will be allotted on a first – come, first served basis.



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